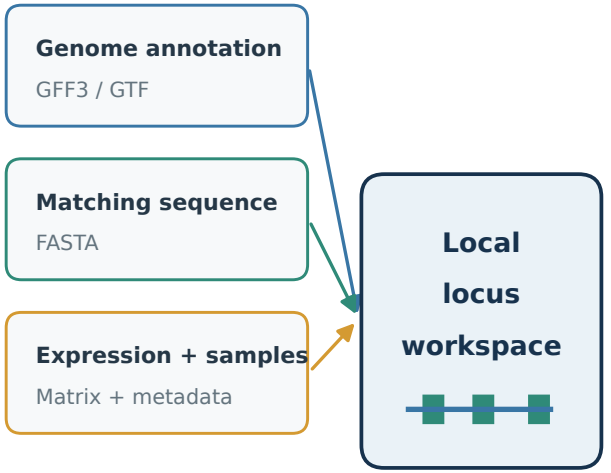
PhytoScope

A portable local query framework for plant genomes

Connect public files to traceable locus-level queries—without requiring a species-specific web database

1 | Bring local data



Search genes · inspect transcript structures

Export sequences · display expression

Save and reopen a portable project

2 | Test portability and fidelity

4,641,758 recognized structural tuples

8 independent source inventories

0 missing · 0 extra

7 plant species evaluated

- Arabidopsis
- Rice
- Maize
- Soybean
- Rice bean
- Andrographis
- Hibiscus

Independent sequence checks

5 coding loci × 3 additional plants

Expression ID join

34,914 / 34,914 IDs

Evidence supports tested releases and formats—not universal compatibility.

3 | Interpret within evidence

Rice-bean expression case

Reads: IC426787 | reference: VRB3

Import / identifier reconciliation only

✓ **Inputs remain authoritative**

No function or orthology inferred

✓ **Missing values stay missing**

Annotation completeness bounds outputs

✓ **Claims stay technical**

No usability, time-saving, or discovery claim

TAKE-HOME **A reusable query layer for researchers working with plant species that lack dedicated online gene portals.**

PhytoScope organizes user-supplied data locally; provenance, reference matching, and annotation completeness define what can be concluded.